Canonicalise What You Can, Measure What You Cannot: Rigid-Body Symmetry in Flow-Matching Motion Planning

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Abstract

Motion planning is exactly equivariant under rigid transformations, and a substantial literature builds this symmetry into network architectures. We show that on a generative planning problem, almost all of the available benefit is obtained without touching the architecture at all — and we give a cheap measurement that says, *before* an equivariant model is built, how much benefit is left for it to capture.

Our starting point is that the start and goal of a planning query themselves define a frame, and that this frame removes five of the six degrees of freedom of SE(3) exactly, at zero computational cost. The sixth — a roll about the start–goal axis — cannot be removed by any continuous rule; we give two independent obstructions, one topological and one distributional. We therefore treat the sixth by symmetry: roll augmentation during training and frame averaging at inference.

On a cluttered 3D point-mass benchmark the reduction raises the held-out collision-free rate from 15.4% to 45.6% (≈ 11 standard errors), and, contrary to the hypothesis that canonicalisation is a small-data crutch, the gap *widens* monotonically with training data: the world-frame baseline gains 2.6 points across a $12.5\times$ increase in training environments while the canonicalised model gains 26.6 and has not plateaued. The two symmetry mechanisms contribute +1.2 and +0.2 points, both inside noise. We explain this rather than merely reporting it, by measuring the *non-equivariance residual*: only 1.6% of the learned velocity field lies outside the equivariant subspace, and that fraction *decreases* with scale. Because frame averaging is the orthogonal projection onto that subspace, 1.6% is the entire budget any post-hoc symmetrisation of this model can spend. We propose the residual as a one-line pre-flight check for equivariant design: measure it first, and build the constrained architecture only if it is large.

I Introduction

A motion planning problem consists of a workspace populated with obstacles, a start configuration $\mathbf{s}$, and a goal configuration $\mathbf{g}$; its solution is a path τ joining them and avoiding the obstacles. The problem carries an exact symmetry: for any rigid transformation $T \in \mathrm{SE}(3)$,

$$\tau\big(T\cdot\mathcal{O}, T\mathbf{s}, T\mathbf{g}\big) \;=\; T\cdot\tau\big(\mathcal{O}, \mathbf{s}, \mathbf{g}\big), \tag{1}$$

where $\mathcal{O}$ denotes the obstacle set. Nothing about the geometry changes when the problem is picked up and moved.

A neural planner trained on raw world coordinates has no access to Eq. (1). As far as its weights are concerned, planning left-to-right near the ceiling is an unrelated task from planning front-to-back near the floor, and the same skill must be relearned at every pose. The dominant response in the literature is architectural: constrain the layers so that equivariance holds by construction [5, 24, 8, 6, 9], an approach that has been carried into robot learning with real success [21, 26, 23, 27]. That success comes at a price that is rarely quantified: constrained layers cost expressiveness, are laborious to implement, and fail in ways that are silent.

There is a second route, older and much cheaper: build the symmetry into the *representation of the problem*, so that the network is never shown the pose and therefore cannot be sensitive to it [18, 11]. This paper asks how far that route goes on a generative motion planning problem, and — more importantly — how one can *know in advance* whether the expensive route is worth taking.

Our answer has three parts.

(i) Five of six degrees of freedom are free. The query supplies a frame. Placing the origin at the midpoint of $\mathbf{s}$ and $\mathbf{g}$ and aligning the first axis with the start–goal direction removes three translational and two rotational degrees of freedom, exactly and at no cost. The resulting representation is *bit-identical* under any rigid transformation of the problem (Proposition 1) — a stronger guarantee than a trained equivariant architecture provides, since it holds to floating-point precision rather than to optimisation tolerance. As a side effect the six numbers previously used to condition the network on the query collapse to a single invariant scalar.

(ii) The sixth is provably not removable, for two unrelated reasons. No continuous rule can fix the roll about the start–goal axis: such a rule is a nowhere-vanishing tangent field on S^2, which the hairy ball theorem forbids (Proposition 2). Deriving the roll from the scene instead also fails, because the obstacle distribution is C_4-symmetric in every coordinate plane and its low-order angular moments vanish identically (Proposition 3). We therefore handle the sixth degree of freedom by symmetry rather than canonicalisation.

(iii) The symmetry machinery is inert here,

and we measure why. Roll augmentation and frame averaging contribute +1.2 and +0.2 percentage points, both inside noise, while the reduction contributes +30.2. Rather than infer inertness from flat curves — a weak conclusion — we measure the quantity frame averaging exists to remove. The *non-equivariance residual*, the disagreement among rotated copies of the velocity field, is 1.6%. Since frame averaging is the orthogonal projection onto the equivariant subspace (Proposition 4), that 1.6% is the complete budget available to any post-hoc symmetrisation of this model (Corollary 5). The residual moreover *shrinks* with training data, from 1.86% to 1.57%: the property an equivariant architecture would impose is arriving on its own and strengthening.

We regard (iii) as the transferable contribution. The residual is a few lines of code, costs one forward pass, requires no retraining, and can be computed on any model with a group action on its inputs. It converts "we tried equivariance and it did not help" — an anecdote — into a quantitative statement about how much headroom a symmetry-based method has on a given problem. We recommend it as a pre-flight check before committing to a constrained architecture.

II Related Work

Equivariant architectures. Group-equivariant convolution [5] generalises translation weight-sharing to larger groups; for SO(3) and SE(3) the standard constructions are tensor field networks [24], SE(3)-Transformers [8], vector neurons [6], and the `e3nn` framework [9]; [2] surveys the area. In robot learning these ideas underpin neural descriptor fields [23], equivariant descriptor fields [21], equivariant diffusion policy [26], and SIM(3)-equivariant policies [27]. This work does not dispute those results; it provides a measurement that indicates when they are the right instrument.

Canonicalisation and frame averaging. Frame averaging [18] obtains equivariance by averaging an unconstrained network over a set of frames derived from the input, and learned canonicalisation [11] predicts a single frame with a small auxiliary network. Both leave the backbone unconstrained. Our $(\mathbf{s}, \mathbf{g})$ reduction is a canonicalisation in this sense, with the distinguishing feature that the frame is not learned or searched but read directly off the query, so five degrees of freedom are removed in closed form; the obstruction of Proposition 2 is precisely the reason the sixth cannot be handled the same way, and is why we fall back to frame averaging there.

Generative motion planning. Diffusion and flow models over whole trajectories have become a standard planning substrate [10, 4, 3, 25]. We use the conditional flow matching / rectified flow formulation [13, 14, 1] for its straight-line interpolants and few-step sampling. Learned planners that regress directly to paths [20, 7] form the non-generative counterpart. To our knowledge, none of this line reports a direct measurement of how far its learned field departs from equivariance.

Classical planners supply our supervision: RRT-Connect [12] for feasibility, then shortcutting and CHOMP [28] refinement, with TrajOpt [22] available in the benchmark.

III Problem Setup

A Benchmark

Experiments use PointMass3D, a benchmark we built for this study. A spherical robot of radius 0.03 moves in the workspace $[-1, 1]^3$ populated with 20 spheres and 20 axis-aligned boxes per environment, obstacle centres drawn uniformly in $[-0.75, 0.75]^3$. Collision checking uses an analytic signed distance field: the environment SDF is the minimum over obstacle SDFs and the workspace walls, and clearance subtracts the robot radius. Start and goal are drawn uniformly subject to a collision margin and a minimum separation $\|\mathbf{g}-\mathbf{s}\| \geq 1.5$. Forty obstacles in a two-unit cube is dense clutter; absolute success rates are correspondingly low and should be read comparatively.

Expert demonstrations come from RRT-Connect, random shortcutting, uniform resampling to 64 waypoints, and CHOMP refinement, with dense collision validation and fallback to the raw RRT path. The dataset is 300 environments × 600 start–goal pairs × 30 trajectories, doubled by time-reversal.

B Flow-matching planner

The planner is a conditional flow-matching model over whole trajectories. A trajectory is $N = 64$ waypoints in $\mathbb{R}^3$. Training draws $\mathbf{x}_0 \sim \mathcal{N}(0, I)$ and $t \sim \mathcal{U}[0, 1]$, forms the interpolant $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\,\mathbf{x}_1$ with $\mathbf{x}_1$ an expert trajectory, and regresses the velocity target:

$$\mathcal{L} = \mathbb{E}_{\mathbf{x}_0,\mathbf{x}_1,t} \left\| v_\theta(\mathbf{x}_t, t, c) - (\mathbf{x}_1 - \mathbf{x}_0) \right\|^2, \qquad (2)$$

where c encodes the obstacles and the query. Sampling integrates $\dot{\mathbf{x}} = v_\theta(\mathbf{x}, t, c)$ from $t = 0$ to 1 by explicit Euler. The network is a dilated temporal convolutional trunk with FiLM conditioning [16, 17] fed by a PointNet-style permutation-invariant obstacle encoder [19]; 2.16M parameters throughout.

IV Method

A Decomposing the group

SE(3) has six degrees of freedom. The query supplies a frame, and that frame consumes five of them:

- **Three translations**, removed by placing the origin at the midpoint $\mathbf{o} = \frac{1}{2}(\mathbf{s} + \mathbf{g})$.
- **Two rotations**, removed by aligning the first axis with $\hat{\mathbf{x}} = (\mathbf{g} - \mathbf{s})/\|\mathbf{g} - \mathbf{s}\|$. Specifying a direction on S^2 costs exactly two degrees of freedom.

- **One rotation remains**: the roll about $\hat{\mathbf{x}}$. Having fixed which way the path runs, nothing in the problem specifies which way is "up" around that axis.

Concretely, let $d = \|\mathbf{g} - \mathbf{s}\|$, let $i^* = \arg\min_i |\hat{\mathbf{x}}_i|$, and set

$$\hat{\mathbf{y}} \propto \hat{\mathbf{x}} \times \mathbf{e}_{i^*}, \qquad \hat{\mathbf{z}} = \hat{\mathbf{x}} \times \hat{\mathbf{y}}, \tag{3}$$

with R the rotation whose rows are $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$. A point $\mathbf{p}$ maps to $R(\mathbf{p} - \mathbf{o})$. Because the smallest component of a unit vector satisfies $|\hat{\mathbf{x}}_{i^*}| \le 1/\sqrt{3}$, the cross product in Eq. (3) has norm $\ge \sqrt{2/3} \approx 0.816$ everywhere, so the construction is well-conditioned over the entire domain with no threshold or fallback branch, and $\det R = +1$ by construction.

Proposition 1 (Exactness)**.** *For every $T = (Q, \mathbf{t}) \in \mathrm{SE}(3)$, the reduced representation of the transformed problem $(T\mathcal{O}, T\mathbf{s}, T\mathbf{g})$ equals that of $(\mathcal{O}, \mathbf{s}, \mathbf{g})$.*

Proof sketch. Under T the midpoint maps to $Q\mathbf{o} + \mathbf{t}$ and the direction to $Q\hat{\mathbf{x}}$. The tie-break index i^* is the only discontinuous element of the construction; away from its measure-zero tie set the frame transforms as $R \mapsto RQ^\top$, so $RQ^\top(Q\mathbf{p} + \mathbf{t} - Q\mathbf{o} - \mathbf{t}) = R(\mathbf{p} - \mathbf{o})$. □

The consequence worth stating plainly: the network never observes the pose, so it cannot be sensitive to it. This is not an approximation that training tightens — it holds identically, to floating-point precision, at initialisation. Two further consequences follow. The six numbers $(\mathbf{s}, \mathbf{g})$ collapse to the single invariant scalar d, and every pair of training examples that were "the same problem in a different pose" now coincide, raising the effective density of the training set without generating a single new trajectory.

B Why the sixth degree of freedom is not removable

Proposition 2 (Topological obstruction)**.** *There is no continuous map $\hat{\mathbf{y}} : S^2 \to S^2$ with $\hat{\mathbf{y}}(\hat{\mathbf{x}}) \perp \hat{\mathbf{x}}$ for all $\hat{\mathbf{x}}$.*

Proof. Such a map is a continuous nowhere-vanishing unit tangent vector field on S^2, whose existence the hairy ball theorem denies [15]. □

Every deterministic gauge therefore carries a discontinuity; one may relocate it but never remove it. The escape route of restricting to a contractible subdomain is closed here: the minimum separation 1.5 along an axis of extent 2.0 still admits axis-aligned start–goal directions, so the domain is the whole sphere. Our i^* tie-break is exactly such a relocated discontinuity, placed on a measure-zero set.

Proposition 3 (Distributional obstruction)**.** *Let obstacle centres be i.i.d. uniform on a cube and let φ_j be their angles in the plane normal to $\hat{\mathbf{x}}$, with weights w_j depending only on radius. Then the angular moments $c_m = \sum_j w_j e^{im\varphi_j}$ satisfy $\mathbb{E}[c_m] = 0$ for all $m \not\equiv 0 \pmod 4$; in particular $\mathbb{E}[c_1] = \mathbb{E}[c_2] = 0$.*

Proof sketch. The cube-uniform law is invariant under the C_4 rotation of any coordinate plane, and a C_4-invariant angular density has vanishing Fourier coefficients except at multiples of 4. □

So a scene-derived gauge — "point $\hat{\mathbf{y}}$ at the obstacle mass" — is not merely occasionally ill-conditioned but degenerate in expectation. Under the null, $|c_1|^2 / \sum_j w_j^2$ is Exp(1) with median ≈ 0.69, so near-degenerate configurations are the typical case rather than a rare tail. The first surviving moment, $m = 4$, is an artefact of cube sampling, is fourfold ambiguous by construction, and would not transfer to a realistic scene distribution. Both routes to a canonical roll are closed, for unrelated reasons.

C Handling the sixth by symmetry

Roll augmentation. During training a uniform $\theta \sim \mathcal{U}[0, 2\pi)$ is composed into R, so reduction and roll are applied as a single rotation. The prior $\mathbf{x}_0$ is drawn *after* this rotation and never rotated: an isotropic Gaussian is already rotation-invariant, so rotating it is a redundant step capable only of introducing error.

Frame averaging. At inference the field is evaluated at K rolled copies of the state, each result rotated back, and averaged:

$$(\mathcal{A}f)(\mathbf{x}) = \frac{1}{K} \sum_{k=0}^{K-1} \rho(\theta_k)^{-1} f\big(\rho(\theta_k)\mathbf{x}\big), \quad \theta_k = \frac{2\pi k}{K}. \tag{4}$$

Proposition 4 (Exact C_K equivariance and projection)**.** *For any f, $\mathcal{A}f$ is exactly C_K-equivariant. Moreover $\mathcal{A}$ is linear, idempotent, and self-adjoint with respect to $\langle f, g \rangle = \mathbb{E}_{\mathbf{x}}\left[f(\mathbf{x})^\top g(\mathbf{x})\right]$ for any C_K-invariant law on $\mathbf{x}$; hence it is the orthogonal projection onto the subspace of C_K-equivariant fields.*

Proof sketch. Rolling the input by θ_j permutes the quadrature points, and reindexing a finite sum is a bijection on them, giving equivariance and $\mathcal{A}^2 = \mathcal{A}$. Self-adjointness follows by substituting $\mathbf{x} \mapsto \rho(\theta_k)\mathbf{x}$ in the expectation, which the invariance of the law leaves unchanged, together with $\rho(\theta_k)^\top = \rho(\theta_k)^{-1}$. □

Two properties of Eq. (4) matter in practice. The guarantee requires no architectural constraint and no training assumption — it is a property of the operator, and holds for an untrained network. And because the obstacle encoding is time-independent, the K rolled scene encodings are computed once outside the ODE loop and reused at every integration step, which is why K=9 costs 1.05× the wall-clock time of K=1 (Sec. V-D).

D The non-equivariance residual

Proposition 4 turns a qualitative question — "is symmetry worth enforcing here?" — into a measurable one. Write $v_k = \rho_k^{-1} f(\rho_k \mathbf{x})$ and $\bar{v} = \mathcal{A}f(\mathbf{x})$, and define

$$r = \frac{\mathbb{E}\|v_k - \bar{v}\|}{\mathbb{E}\|\bar{v}\|}. \tag{5}$$

This is the relative norm of the component of the learned field that lies outside the equivariant subspace.

Corollary 5 (Symmetrisation budget)**.** *The orthogonal decomposition $f = \mathcal{A}f + (I - \mathcal{A})f$ gives $\|f - \mathcal{A}f\|/\|f\| = r/\sqrt{1+r^2}$, and $\mathcal{A}f$ is the closest C_K-equivariant field to f in $\|\cdot\|$. Hence any procedure that post-processes f into an equivariant field changes it by at most a relative r, and the remaining error of f is equivariant error.*

The practical reading is a decision rule. Measure r; if it is small, the model's error is not a symmetry violation, and symmetrising it — by frame averaging, by augmentation, or by any other post-hoc projection — cannot remove more than r of the field. The measurement costs one forward pass and no retraining.

We state the limit of this argument explicitly, because it is the objection a reader should raise. Corollary 5 bounds what can be gained by symmetrising *this* model. It does not bound what an equivariant architecture could achieve, since such an architecture searches a different hypothesis class and may converge to a different — possibly better — equivariant field. What a small r does establish is that the model's residual error is overwhelmingly *within* the equivariant subspace, so an architecture that merely enforces membership of that subspace is addressing a constraint that is already satisfied.

E Implementation

Three changes make the reduction representable, and one class of error had to be designed out.

Oriented boxes. The reduction is a general SO(3) rotation, under which an axis-aligned box becomes oriented. The usual six-number representation (centre, half-extents) cannot express this. We use twelve numbers: a centre and three half-edge *vectors*, which in the world frame are $(h_x, 0, 0)$, $(0, h_y, 0)$, $(0, 0, h_z)$, so the re-encoding is lossless. The environment SDF is unchanged, since collision checking happens in the world frame after the inverse transform.

Points and free vectors transform differently. This is the error the implementation is principally designed to prevent, because it fails silently. Waypoints, endpoints, and obstacle centres are *points* and take the affine map $R(\mathbf{p} - \mathbf{o})$; the box half-edges are *free vectors* and take the rotation only, $R\mathbf{v}$. Applying the affine map to a half-edge adds $-R\mathbf{o}$ to every edge — identically — so box *position* survives while box *extent* is swamped by a common offset. The model goes blind to obstacle size while the loss curve stays healthy. The severity scales with $\|\mathbf{o}\|$, so the defect is invisible on exactly those problems whose endpoints straddle the origin.

Verification. Eighteen tests cover deliberately disjoint failure classes. Asserting $\mathbf{s}, \mathbf{g} \mapsto (\mp d/2, 0, 0)$ catches translation errors, axis ordering, and R versus $R^\top$, but is blind to handedness, since a reflection in the y–z plane fixes the x-axis pointwise; $\det R > 0$ covers that. Neither detects the point/vector confusion, so a third test asserts featurise$(\rho \cdot S)$ = typed_rotate(featurise$(S), \rho)$ with the reference side written as explicit loops rather than the batched operations under test, so it catches indexing and reshaping errors instead of agreeing with them. A separate suite verifies Eq. (4) against a deliberately *untrained* network — the hard case, since Proposition 4 should not depend on f.

One subtlety is worth recording for anyone measuring r. The group acts on the whole input, scene and state together. An early version of our equivariance test rolled only the state while leaving the conditioning in its original frame, and reported a large violation from a correct implementation. A diagnostic written that way manufactures the non-equivariance it claims to find.

V Experiments

A Protocol

All results use environments 250–299, which no model was trained on, evaluated as 50 environments × 10 start–goal pairs × 20 samples with 8 Euler steps. The two arms are:

- **Control**: world frame, conditioned on the raw pair $(\mathbf{s}, \mathbf{g})$.
- **Treatment**: the $(\mathbf{s}, \mathbf{g})$ reduction with uniform roll augmentation, conditioned on the invariant scalar d.

Both arms use the twelve-dimensional oriented-box representation, so it is not a confound; both train on identical trajectories for 20 epochs.

Validation split. Within training environments, validation holds out whole start–goal pairs. A per-trajectory split leaks badly here: the 30 trajectories per pair are near-duplicates (within-pair standard deviation 0.054 against an overall 0.426, because CHOMP is a local optimiser and converges to the same route from similar initialisations). Under a per-trajectory split a held-out trajectory has a training neighbour at RMS distance 0.030; under the pair split this rises to 0.162.

Statistics. We treat the number of *distinct problems* (500) as the sample size rather than the number of trajectories (10,000), since samples within a problem and problems within an environment are correlated. This gives a standard error of ≈ 2 percentage points, and ≈ 2.7 for a difference of two arms.

Loss is not comparable across arms. The treatment regresses targets in the reduced frame, where trajectories are canonicalised along $+\hat{\mathbf{x}}$, so its target variance and hence its MSE are mechanically smaller. All reported quantities are world-frame task metrics.

B The reduction scales; the baseline does not

Table 1 and Fig. 1 give the principal result. At 250 environments the reduction reaches 45.6% against 15.4%, a gap of 30.2 points, or roughly 11 standard errors.

Training envs	Collision-free (%) Control	Ours	Min. clearance Control	Ours
20	12.8	19.0	−0.0776	−0.0583
60	13.5	31.3	−0.0746	−0.0312
150	14.6	41.0	−0.0726	−0.0163
250	15.4	**45.6**	−0.0697	**−0.0095**

Table 1. Held-out performance against training-set size. The control gains 2.6 points across a 12.5× increase in environments; the reduction gains 26.6 and has not plateaued. Minimum clearance is an independent, continuous measure and tells the same story

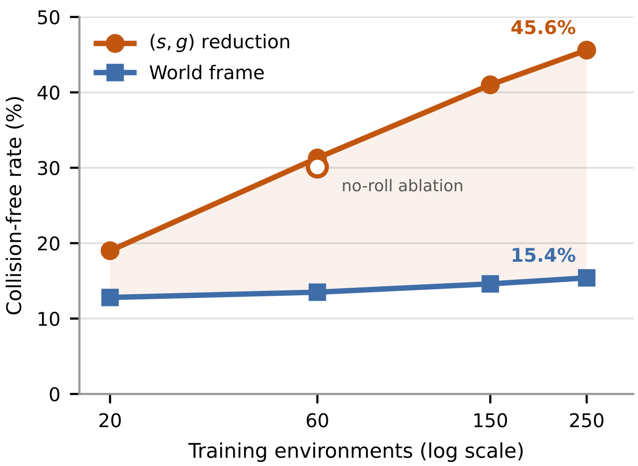

Figure 1. The gap widens with data. If canonicalisation were compensating for a shortage of training data one would expect it to close; instead the world-frame baseline is nearly flat while the reduced model continues to improve. The hollow marker is the no-roll ablation, drawn on the treatment line because it is the same arm with one component removed

The trend is more informative than the endpoint. The baseline improves by 2.6 points across a 12.5× increase in training environments: it is close to unable to exploit additional layouts, because each new layout arrives at poses it must learn about separately. The reduction improves by 26.6 points over the same range and has not levelled off. This is the opposite of what a small-data-crutch account predicts, under which the advantage should shrink as data accumulates. Minimum clearance corroborates through an independent continuous measure: the treatment moves from −0.058 toward −0.0095, approaching feasibility, while the baseline is static.

Endpoint error falls by a factor of three to five under the reduction (0.0101 → 0.0041 at 20 environments, 0.0059 → 0.0020 at 250), a direct consequence of canonicalisation: with endpoints always at $(\mp d/2, 0, 0)$ the model no longer spends capacity on *where* they are.

C The mechanisms for the sixth degree of freedom are inert

Table 2 and Fig. 2a decompose the result. Removing roll augmentation (the reduction with a fixed gauge) costs 31.3% → 30.1% at 60 environments. Frame averaging on

Mechanism	Δ free	Assessment
$(\mathbf{s}, \mathbf{g})$ reduction (5 DOF)	+30.2 pp	≈ 11 SE; grows
Roll augmentation	+1.2 pp	0.4 SE (noise)
Frame averaging (K:1 → 9)	+0.2 pp	0.09 SE (≈ 0)

Table 2. Decomposition by mechanism on the same held-out set. The entire effect is attributable to canonicalisation

the best model gives 45.6/45.7/45.8% for $K = 1, 3, 9$ — 0.09 standard errors.

Rather than infer inertness from flat curves, we measure r. On the best model $r = 0.0157$ at K=3 and 0.0174 at K=9: under 2% of the field is non-equivariant. By Corollary 5 that is the whole budget, and the remaining error — with the model at 45.6% collision-free, there is a great deal of it — respects the symmetry and is untouchable by symmetrisation.

Two observations sharpen this. First, the residual barely moves when roll augmentation is removed (0.0186 → 0.0292 at 60 environments, both small), so augmentation did not cause the near-equivariance. The explanation is that the gauge is a deterministic function of $\hat{\mathbf{x}}$, and 600 start–goal pairs per environment supply 600 distinct directions; the reduced-frame data already spans a wide spread of rolls, and explicit augmentation was solving a problem the data had already solved. Second, the residual *falls* with scale, from 0.0186 at 60 environments to 0.0157 at 250 (Fig. 2b): the model becomes more equivariant as it sees more layouts.

The second observation is the strongest available argument against building the constrained architecture for this problem. The property such an architecture would enforce by construction is emerging on its own and strengthening with data, so imposing it would trade expressiveness for something obtained free of charge.

D Cost

Frame averaging at K=9 costs 1.05× the wall-clock time of K=1 (0.070 vs 0.067 s per query). At 64 waypoints and batch size 1 the sampler is bound by kernel launch overhead rather than arithmetic, so a ninefold increase in width is nearly free. This matters for the interpretation: the finding that frame averaging does not help is not a budget trade-off. It was affordable, and we adopted it anyway for the measurement.

Separately, reducing the integrator from 20 to 8 Euler steps costs 0.4–0.8 points and runs 1.4× faster, so 8 steps is used throughout.

VI Discussion and Limitations

What the technique contributes. Of the six degrees of freedom of SE(3), five can be removed exactly by a frame read off the query, at no computational cost and with no architectural constraint. Doing so roughly triples the collision-free rate on this benchmark and, more importantly, converts a model that cannot use additional training environments into one that can. The sixth cannot be removed and, here, does not need to be

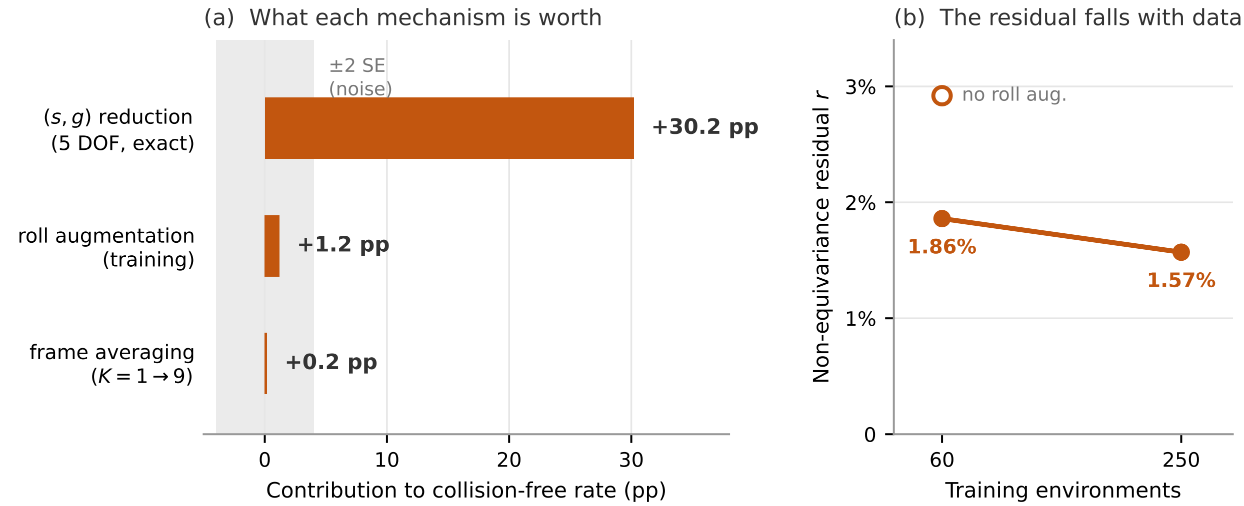

Figure 2. (a) Only the canonicalisation clears the noise floor; the two mechanisms built for the sixth degree of freedom sit inside it. (b) The non-equivariance residual of Eq. (5) falls as training data grows, so the property an equivariant architecture would impose is emerging on its own; the hollow marker shows that removing roll augmentation raises the residual but leaves it small, so augmentation was not the cause of the near-equivariance

— a contingent empirical fact rather than a theorem, but one supported by three independent measurements: the ablation, the K sweep, and the residual.

Absolute performance is poor. Forty obstacles in a two-unit cube is dense, and the model is small (2.16M parameters) and briefly trained (20 epochs). No train–validation gap appeared in any run, indicating underfitting. The comparison between arms is sound — identical data, identical budget, identical obstacle representation — but the level indicates substantial headroom in capacity and training budget, and we do not claim a state-of-the-art planner.

Scope of the negative result. The claim that the sixth degree of freedom needs no treatment is specific to this setting, and relies on the training distribution supplying implicit roll diversity through many start–goal directions. A dataset with few directions per environment, or a task whose angular structure is genuinely richer — manipulation with a gripper, or non-holonomic dynamics — could plausibly reverse it. That is precisely why we advocate measuring r rather than generalising our answer: the diagnostic transfers even where the conclusion does not.

What the residual does and does not bound. As stated in Sec. IV-D, Corollary 5 only bounds post-hoc symmetrisation of a given model, not the achievable performance of a differently-parameterised equivariant model. A conclusive test would train an SO(2)-equivariant backbone and compare; our argument is that a small and shrinking r makes that experiment low-yield, not that it is uninformative.

One honest asterisk. Frame averaging *does* consistently improve endpoint error, by roughly 35% ($0.0020 \to 0.0013$), on every checkpoint tested. This is a real effect. It simply does not propagate to the collision-free rate, because endpoint precision was not the failure mode.

Dataset diversity. The dataset is less diverse than its nominal size suggests: 30 near-duplicate trajectories per pair means 10.8M paths carry on the order of 360k distinct queries. This does not affect the comparison, since both arms train on identical data, but it bears on how additional data should be generated — cutting trajectories per pair from 30 to 5 would buy roughly six times the environments for the same generation cost, and Fig. 1 says environments are the binding constraint.

Single benchmark. Results are from one workspace family and one planner architecture. We report the reduction and the diagnostic in enough detail to be reimplemented elsewhere; whether the 30-point gap survives in a manipulator configuration space is open.

Future work. Beyond more environments and capacity, one option is now open that was previously excluded. The flow begins from an isotropic Gaussian that knows nothing of the query, so it must learn both to produce a path and to place its endpoints. A Brownian-bridge prior pinned at start and goal would leave only the obstacle-avoidance detour to be learned. We had excluded it because such a prior is not isotropic in the plane normal to $\hat{\mathbf{x}}$ and would break the distributional equivariance guarantee at $t = 0$; since that guarantee has now been measured to be worth 0.2 points, the trade is available.

VII Conclusion

The model presented here is exactly SE(3)-equivariant by construction for five of six degrees of freedom, and measured to be 98% equivariant for the sixth without any mechanism enforcing it. All of the measured benefit lies in the five.

The recommendation is correspondingly simple, and we believe it applies beyond this benchmark: canonicalise the degrees of freedom that admit a continuous canonical form — the query usually supplies more of them than one expects — and then *measure* the residual non-equivariance before building machinery for the remainder. The measurement costs one forward pass. On this problem it would have saved the effort of an equivariant architecture, and it said so before any of that effort was spent.

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